

WOVENLOG: VALIDATING EMOTIONALLY GUIDED STORIES WITH LOGIC-BASED REASONING

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INTRODUCTION

Woven is an interactive storytelling platform that guides readers from their current emotional state to a desired target mood through dynamically generated narratives.

- Short, dynamic stories that move you from your current mood to a chosen target emotion
- You enter your mood and pick a story choice
- ChatGPT or Gemini generates the next snippet
- A Python validator maps each snippet to 10 core emotions, enforces transition rules, and logs any mismatches

ALEX’S ADVENTURE STORY

The wind whipped at Alex's face as he stared out at the grey expanse of the ocean. He was supposed to be meeting his friend, Liam, at the pier, but Liam was late. Again. Alex pulled his jacket tighter, the familiar weight a small comfort against the growing chill. The salt spray stung his cheeks, mirroring the sting in his chest. He felt a familiar ache, a dull sadness that had been his constant companion lately.

He saw a small, colorful kite snagged on a lamppost further down the pier. It was swaying back and forth in the wind. Should he go see if he can help?

| What would you do?

Example story prompt

METHOD

Woven takes in the inputs from the users as shown in the image. That information is then stored in state of streamlit and used to create prompts for ChatGPT or OpenAI, based on what the user chose. The prompt is designed to update based on:

- 1. **User Personality scores:** Given by the LLM Model
- 2. **Turn:** If it’s the 5th prompt, the question is more intense

We run each generated turn through our Emotional Validator, which assigns one of ten emotions via an NLP classifier. It applies rules—allowing any shift in turns 1-3, limiting adjacent emotions in turns 4-6, then enforcing strict target alignment—and flags inconsistencies, logging errors without pausing flow.

NAME

AGE

+

-

PRONOUNS

GENRE

CURRENT EMOTION

TARGET EMOTION

MODEL

ChatGPT

Gemini

LENGTH

SUBMIT

Your Story Personality

Risk taker

1/5

Optimism

4/5

Social

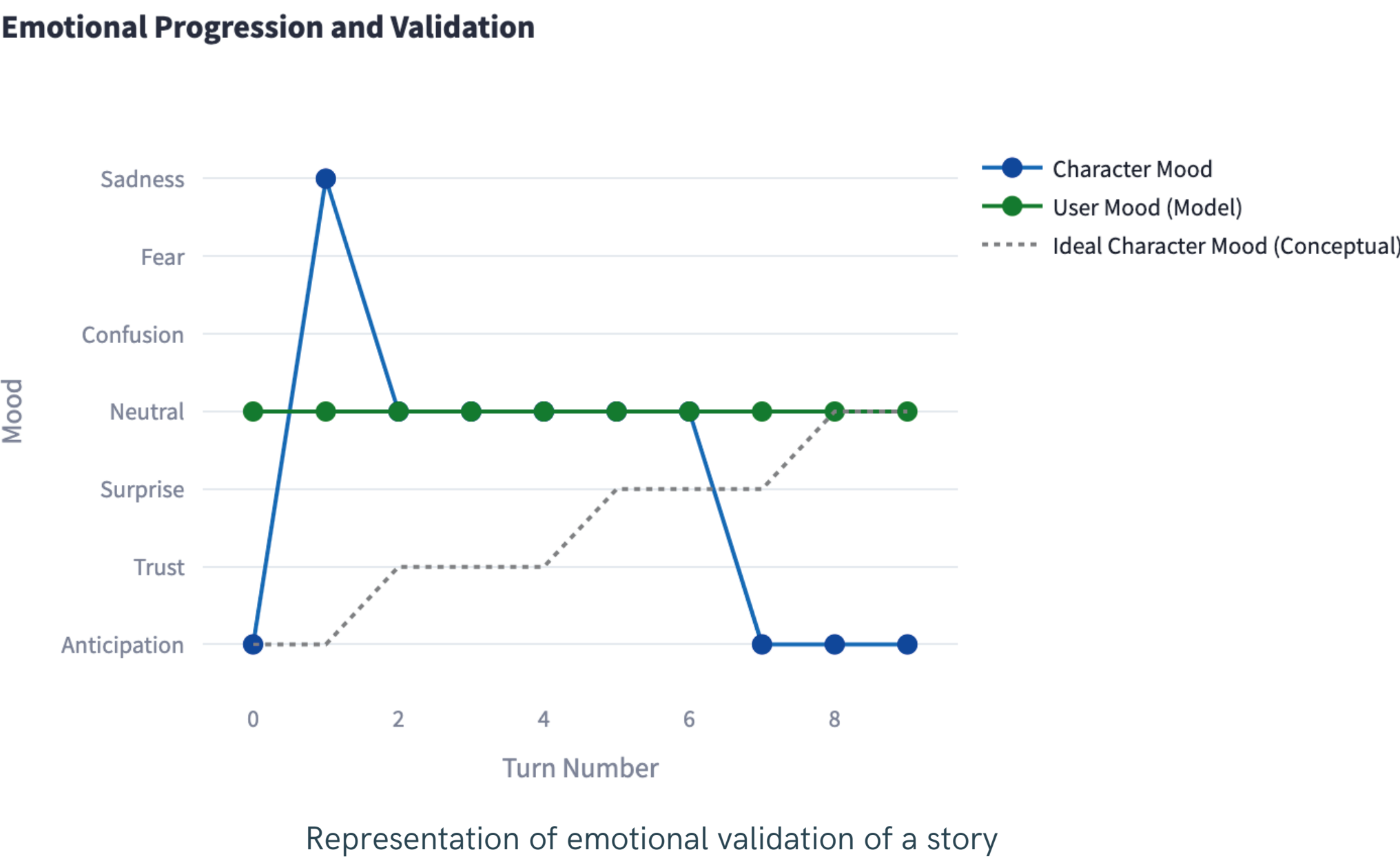
5/5

Analytical

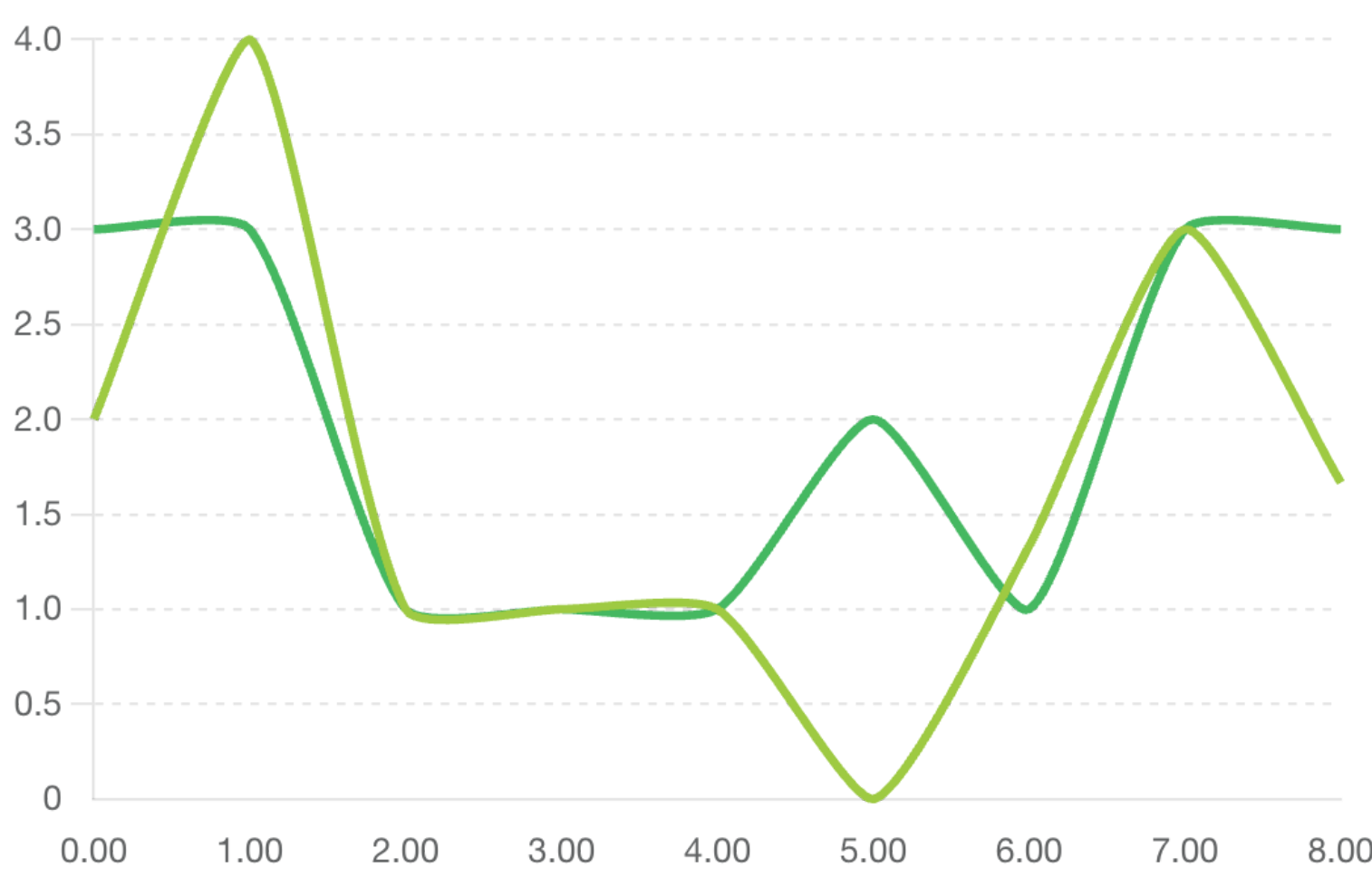
-2/5

Introspective

0/5



RESULT



The average of the three Gemini runs stays within two mood-indices of the ideal on most turns and even hits an exact match at turn 5. OpenAI’s single run, by contrast, never reaches the target, remaining one to three indices off at every step. This comparison shows that Gemini, when averaged across sessions, more faithfully reproduces the intended emotional arc than the OpenAI output.

CONCLUSION AND FUTURE WORK

Woven integrates LLM storytelling, real-time emotional validation, and reader-driven mood inputs to keep narratives coherent and engaging. By tracking alignment hotspots, we continuously refine the story loop. Next steps include:

- **Visual Emotion Timeline:** Add live, turn-by-turn mood graphs and mood-matched illustrations to deepen user connection.
- **UI Enhancements:** Introduce intuitive controls (sliders, drag-and-drop) and responsive layouts for smoother mood selection.
- **Genre-Aware Validation:** Tailor EmotionalValidator rules to specific genres (e.g., rising tension in thrillers, warmth in romance) for richer narrative coherence.